

Coding & Solution

Date	24 September 2026
Team ID	SWTID-2026-2311
Project Name	AI StudyBuddy API
Maximum Marks	8 Marks

Coding & Solution - Implementation Snapshot

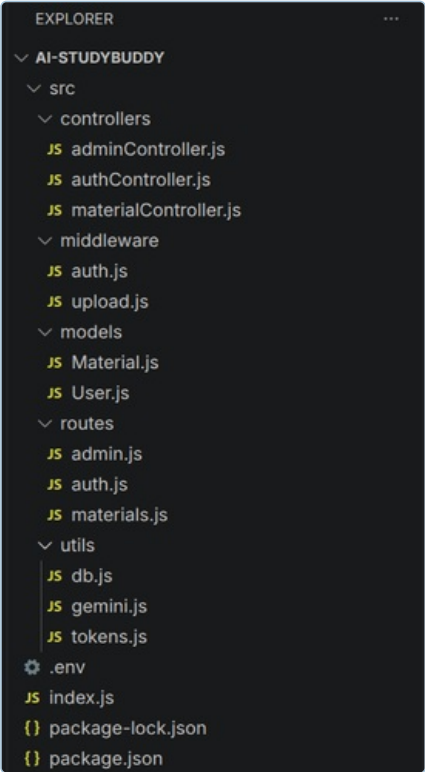


Fig. 1 – Actual project folder structure (VS Code Explorer)

Full source is version-controlled and available at https://github.com/sachinselva038-nm/studybuddy_selva. The backend runs on Node.js/Express with MongoDB Atlas (Mongoose) and exposes Auth, Materials and Admin route groups, with all AI-generation logic nested under the Materials route group.

Representative Endpoint (Upload Study Material)

```
POST /api/materials/upload
Headers: Authorization: Bearer <accessToken>
Body (form-data):
  file: Node js Introduction.txt
  title: Node.Js

Response: 201 Created
{
  "message": "Material uploaded!",
  "material": {
    "user": "aa3f69ab173d7853fc8b3841",
    "title": "Node.Js",
    "content": "Node.js\nIntroduction\n\nNode.js is an open-source, cross-platform\nJavaScript runtime environment ..."
  }
}
```

The `upload.js` middleware (multer) rejects any file outside `.txt / .md / .pdf` and enforces a 5 MB size limit before the controller ever runs – invalid uploads never reach the database.